

Careless/Insufficient Effort Responding (C/IER) Detection

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This notebook presents a tutorial on the detection of careless or insufficient effort responding (C/IER) in self-report questionnaires. We use real database from WHOQOL-Bref scale as an example ($n = 1,546$), and we apply the ‘careless’ package and Laz.R (Biemann et al., 2025) to detect C/IER (post hoc indicators). We also discuss the importance of procedural C/IER detection in the quality control of data obtained through self-report questionnaires.

1 Contextualization

The detection of Careless or Insufficient Effort Responding (C/IER) is an essential step in the quality control of data obtained through self-report questionnaires. C/IER occurs when respondents complete the instrument without attention to the item content, compromising the validity of the psychometric and substantive inferences derived from the data (Curran, 2016; Huang et al., 2026).

The analytical flow of this document follows two sequential levels of detection:

1. **Questionnaire:** procedural exclusions (a priori), including dropout, response time, and quality control items.
2. **WHOQOL Scale:** diagnostic post-hoc indicators (Laz.R, Longstring, Mahalanobis Distance, and IRV) with cut-off points estimated via the Kneedle algorithm (Biemann et al., 2025).

2 Data reading and preparation

The `data.csv` file contains the raw survey responses (separator `;`). In this step, dropout records are removed, variables for the analytical flow are selected, and the total response time (`TIME`) is derived.

Rows: 1,326

Columns: 33

```
$ ID      <dbl> 10441016017, 10440981694, 10440601239, 10440562981, 104403585...
$ CREATED <chr> "28/12/2018 12:00", "28/12/2018 11:25", "28/12/2018 03:23", "...
$ MODIFIED <chr> "28/12/2018 12:20", "28/12/2018 11:47", "28/12/2018 03:39", "...
$ CQ1     <dbl> 1, 1, 1, 1, 1, 1, 1, 1, NA, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ...
$ CQ2     <dbl> 1, 1, 1, 1, 1, 1, 1, 1, NA, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ...
$ CQ3     <dbl> 1, 1, 1, 1, 1, 1, 1, 1, NA, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ...
$ Q1      <dbl> 5, 3, 1, 4, 3, 4, 3, 4, 2, 5, 4, 3, 4, 2, 4, 5, 4, 5, 4, 4, 4...
$ Q2      <dbl> 5, 4, 1, 4, 4, 4, 4, 4, 2, 5, 4, NA, 4, 1, 4, 5, 4, 4, 4, 4, ...
$ Q3      <dbl> 1, 2, 3, 2, 1, 1, 1, 1, 4, 2, 3, 2, 2, 3, 3, 1, 2, 2, 2, 3, 1...
$ Q4      <dbl> 1, 2, 2, 1, 1, 1, 2, 1, 4, 1, 3, 2, 2, 2, 2, 2, 2, 1, 2, 1, 1...
$ Q5      <dbl> 3, 3, 1, 3, 3, 3, 3, 3, 1, 4, 3, 4, 4, 1, 3, 4, 3, 5, 3, 4, 3...
$ Q6      <dbl> 4, 4, 1, 3, 5, 4, 3, 4, 4, 4, 5, 3, 3, 4, 4, 5, 3, 5, 4, 4, 3...
$ Q7      <dbl> 4, 3, 1, 3, 5, 4, 2, 3, 2, 4, 3, 2, 4, 2, 3, 4, 3, 4, 2, 3, 4...
$ Q8      <dbl> 1, 2, 1, 2, 2, 4, 4, 4, 2, 3, 2, 3, 3, 3, 2, 2, 4, 3, 3, 3, 2...
$ Q9      <dbl> 5, 4, 1, 2, 3, 4, 3, 4, 3, 3, 3, 3, 4, 1, 2, 3, 4, 4, 2, 3, 2...
$ Q10     <dbl> 5, 3, 1, 3, 5, 4, 2, 4, 2, 4, 4, 2, 3, 2, 3, 5, 4, 5, 3, 4, 4...
$ Q11     <dbl> 3, 3, 2, 3, 3, 4, 2, 3, 3, 5, 4, 2, 4, 1, 3, 5, 3, 5, 3, 4, 4...
$ Q12     <dbl> 5, 3, 1, 3, 3, 3, 3, 3, 2, 5, 3, 3, 3, 3, 3, 4, 4, 4, 3, 4, 4...
$ Q13     <dbl> 5, 3, 2, 4, 3, 3, 4, 4, 3, 5, 3, 3, 4, 3, 3, 5, 3, 4, 2, 4, 3...
$ Q14     <dbl> 4, 3, 1, 4, 2, 3, 3, 5, 2, 4, 4, 4, 3, 4, 3, 5, 3, 3, 2, 4, 3...
$ Q15     <dbl> 5, 4, 1, 5, 5, 5, 5, 5, 1, 5, 5, 3, 4, 4, 3, 5, 4, 5, 4, 4, 4...
$ Q16     <dbl> 5, 3, 5, 5, 5, 4, 3, 3, 2, 5, 3, 2, 5, 2, 2, 5, 4, 3, 4, 4, 3...
$ Q17     <dbl> 4, 3, 5, 5, 5, 4, 4, 5, 2, 5, 4, 2, 4, 2, 3, 5, 4, 4, 4, 4, 4...
$ Q18     <dbl> 4, 3, 5, 5, 5, 4, 3, 4, 2, 5, 4, 2, 4, 2, 3, 5, 4, 5, 4, 4, 4...
$ Q19     <dbl> 4, 4, 5, 5, 5, 4, 2, 3, 3, 5, 4, 2, 4, 2, 3, 5, 3, 4, 5, 5, 3...
$ Q20     <dbl> 2, 3, 3, 5, 5, 4, 4, 4, 3, 4, 4, 3, 3, 3, 2, 5, 3, 4, 3, 5, 3...
$ Q21     <dbl> 2, 3, 2, 5, 5, 4, 2, 3, 2, 4, 2, 4, 4, 2, 2, 5, 3, 4, 4, 4, 4...
$ Q22     <dbl> 5, 3, 3, 5, 5, 4, 2, 3, 3, 4, 4, 3, 2, 3, 2, 4, 4, 4, 3, 5, 3...
$ Q23     <dbl> 5, 3, 3, 5, 2, 4, 2, 4, 3, 5, 5, 3, 4, 4, 4, 4, 4, 5, 4, 5, 4...
$ Q24     <dbl> 2, 2, 3, 4, 2, 4, 2, 5, 2, 4, 2, 3, 2, 4, 4, 5, 4, 3, 4, 4, 4...
$ Q25     <dbl> 2, 2, 3, 4, 3, 4, 2, 5, 3, 4, 3, 3, 3, 3, 2, 5, 4, 4, 4, 4, 3...
$ Q26     <dbl> 2, 2, 4, 1, 1, 2, 2, 3, 1, 2, 2, 3, 1, 2, 3, 2, 2, 2, 3, 1, 2...
$ TIME    <dbl> 1200, 1320, 960, 1260, 1140, 1140, 540, 1080, 1380, 660, 1620...
```

Source: [Procedural exclusions](#)

3 Procedural exclusions

Level 1 applies three procedures prior to calculating post-hoc indicators: dropout removal, exclusion for response time below 208 seconds ($104 \text{ items} \times 2 \text{ s/item}$) (Huang et al., 2026), and exclusion for two or more failures on quality control items (CQ1–CQ3) (Curran, 2016). Every exclusion is recorded in `exclusion_log.csv` with its reason and criterion.

Table 1: Procedural exclusions summary

Step	N
Raw data	1546
Dropouts removed	220
Excluded by time ($< 208 \text{ s}$)	0
Excluded by QC (≥ 2 fails)	28
Excluded by missings (> 5 in Q1-Q26)	3
Eligible for Level 2	1295
Remaining NAs in Q1-Q26	90

Source: [Procedural exclusions](#)

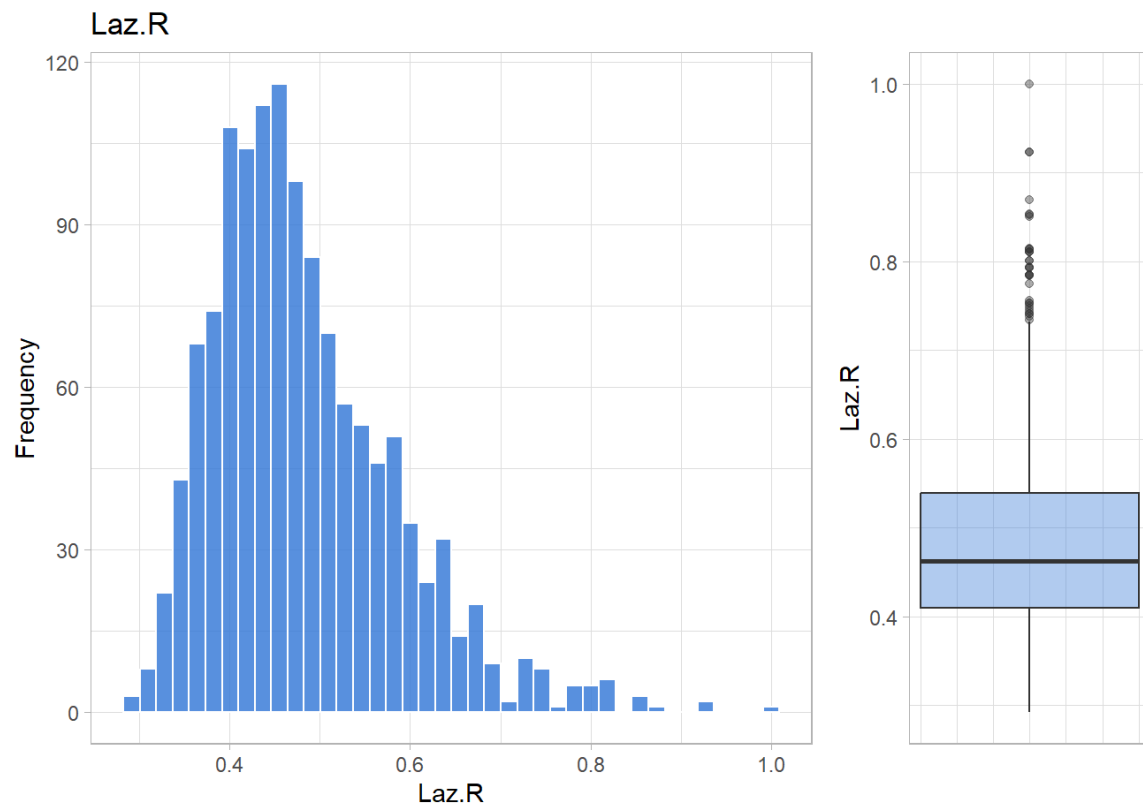
4 Calculation of C/IER indicators

Each indicator captures distinct manifestations of C/IER; their concurrent presence offers stronger evidence than any single indicator alone (Curran, 2016; Hong et al., 2020; Huang et al., 2026).

4.1 Univariate visualization

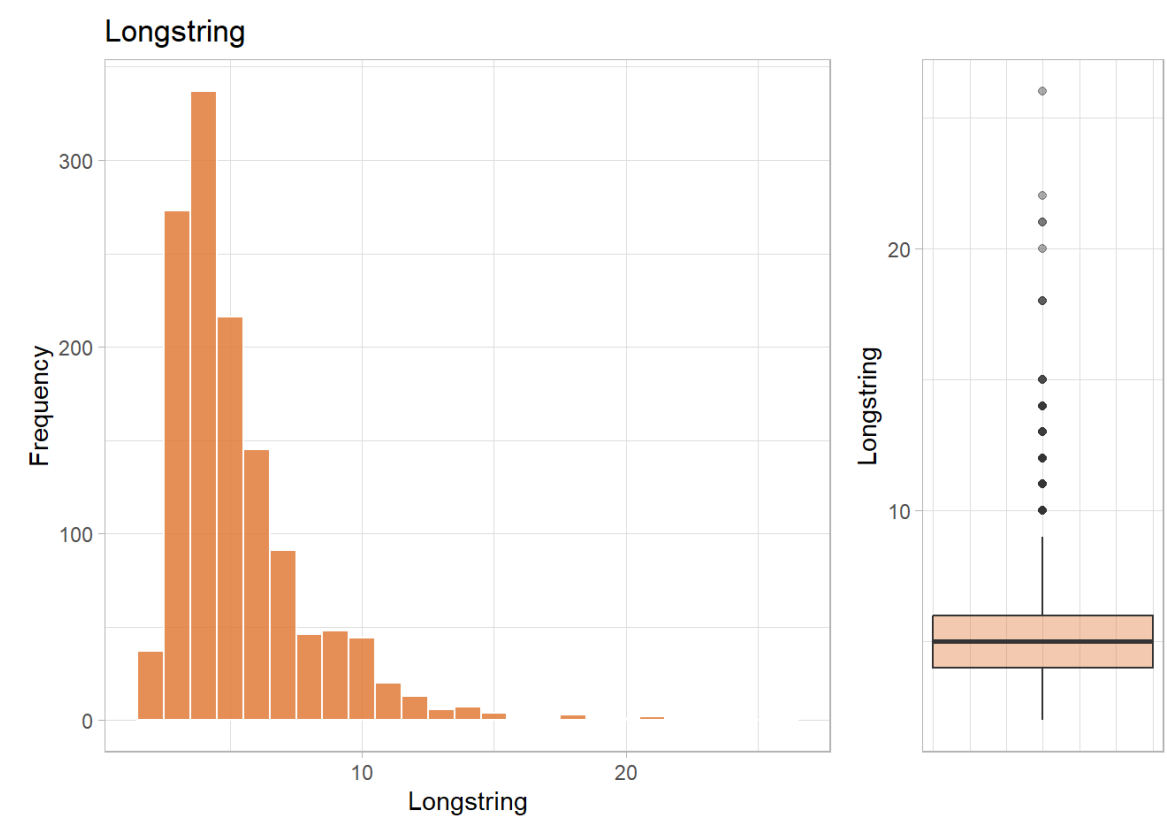
Each respondent receives the values of the four indicators in a consolidated tibble, and the plots below show the univariate distribution of each one.

4.1.1 Laz.R



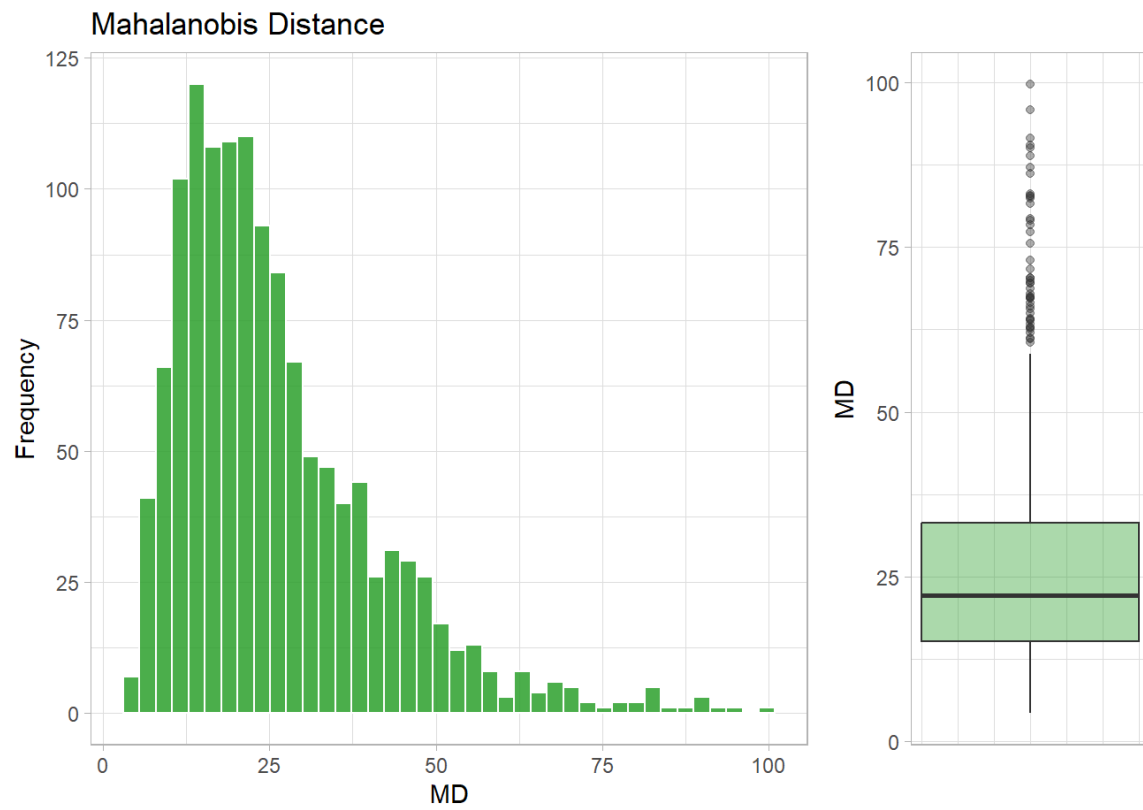
Source: [Post-hoc C/IER Indicators](#)

4.1.2 Longstring



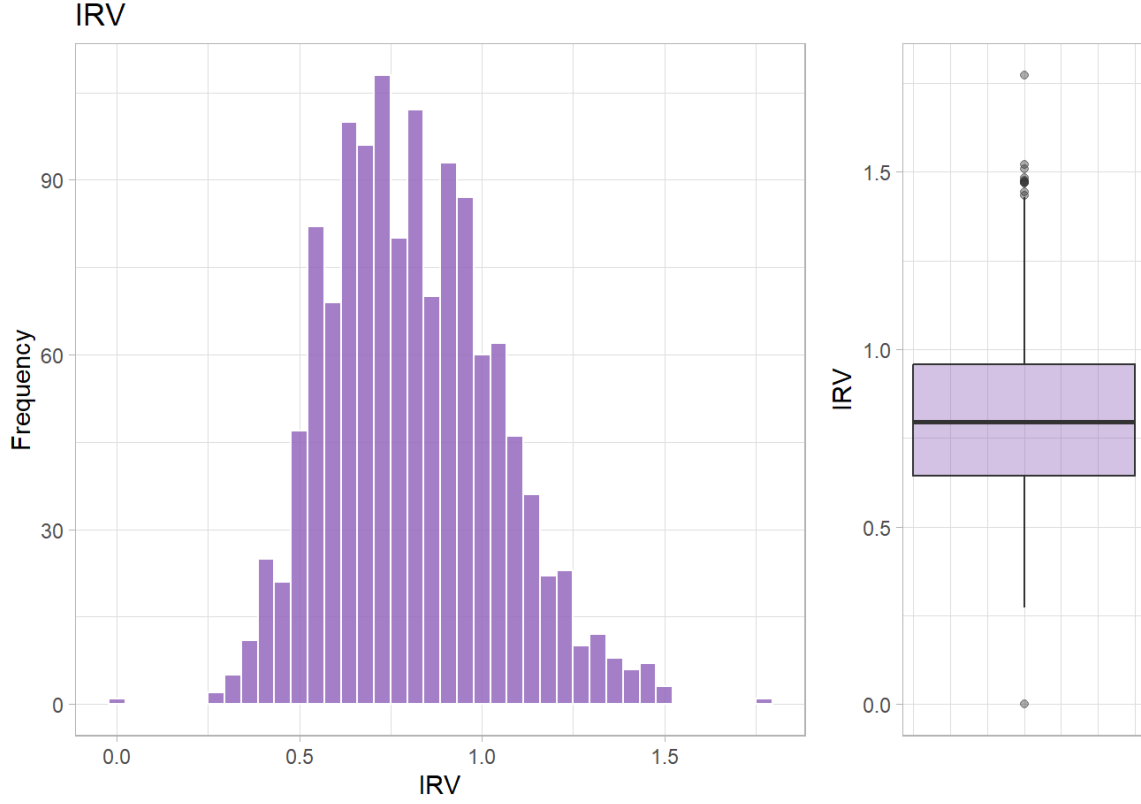
Source: [Post-hoc C/IER Indicators](#)

4.1.3 Mahalanobis Distance



Source: [Post-hoc C/IER Indicators](#)

4.1.4 IRV



Source: [Post-hoc C/IER Indicators](#)

4.2 Cut-off points via Kneedle algorithm

The Kneedle algorithm is applied to the sorted (descending) curve of each indicator. For **Laz.R**, **Longstring**, and **MD**, kneedle with **sign** = -1 identifies where extremely high values separate from the majority (right tail). For **IRV**, **sign** = -1 is used to detect excessively high IRV and **sign** = +1 for excessively low IRV, since both extremes can be considered suspicious.

Table 2: Cut-off points — Kneedle algorithm (Biemann et al., 2025)

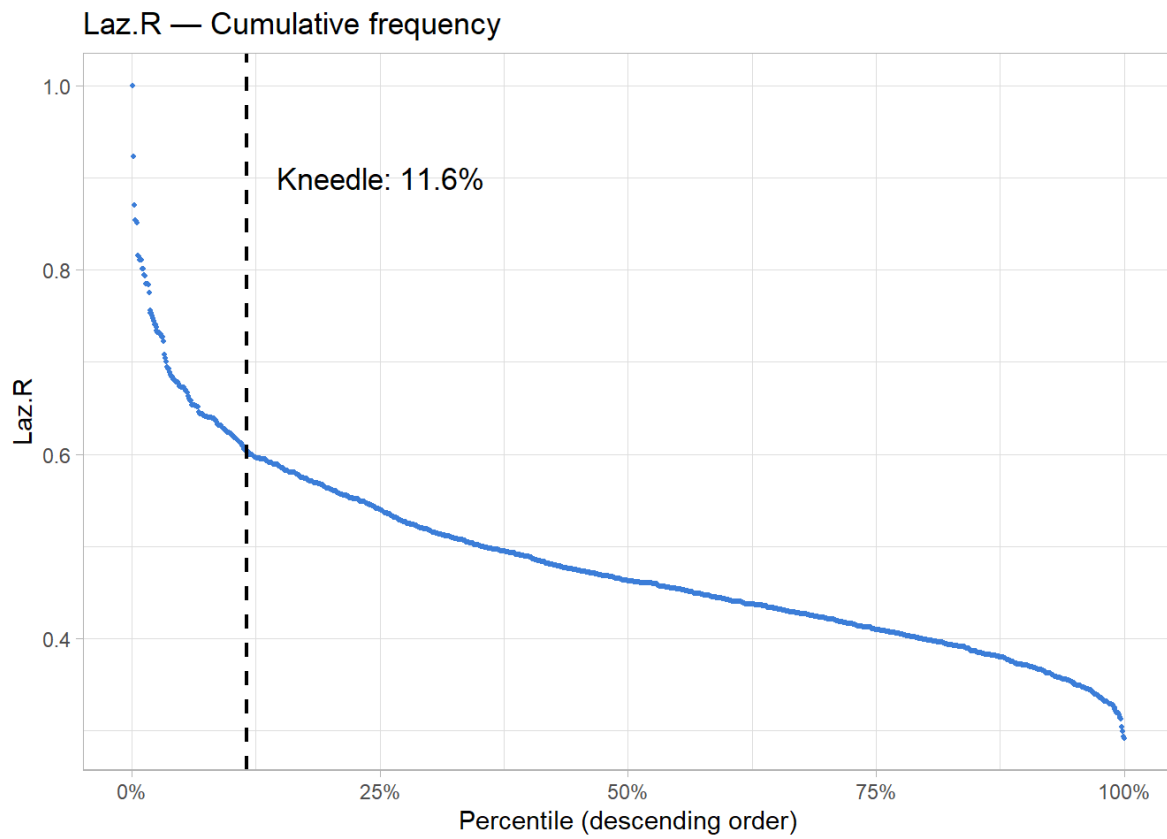
Indicator	Critical value	% flagged
Laz.R	0.60	11.6
Longstring	7.00	15.2
MD	40.12	15.4
IRV (high end)	1.14	8.0

Indicator	Critical value	% flagged
IRV (low end)	0.48	5.0

Source: [Post-hoc C/IER Indicators](#)

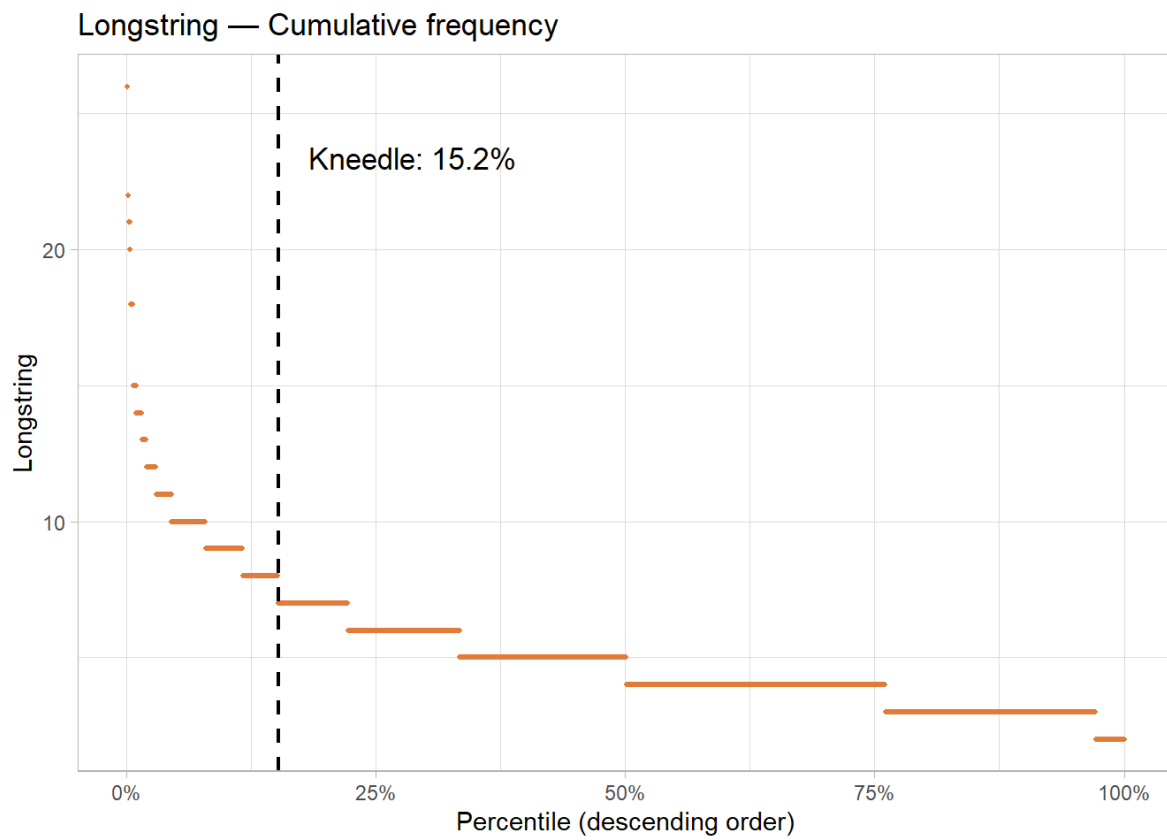
4.3 Plots with cut-off points

4.3.1 Laz.R



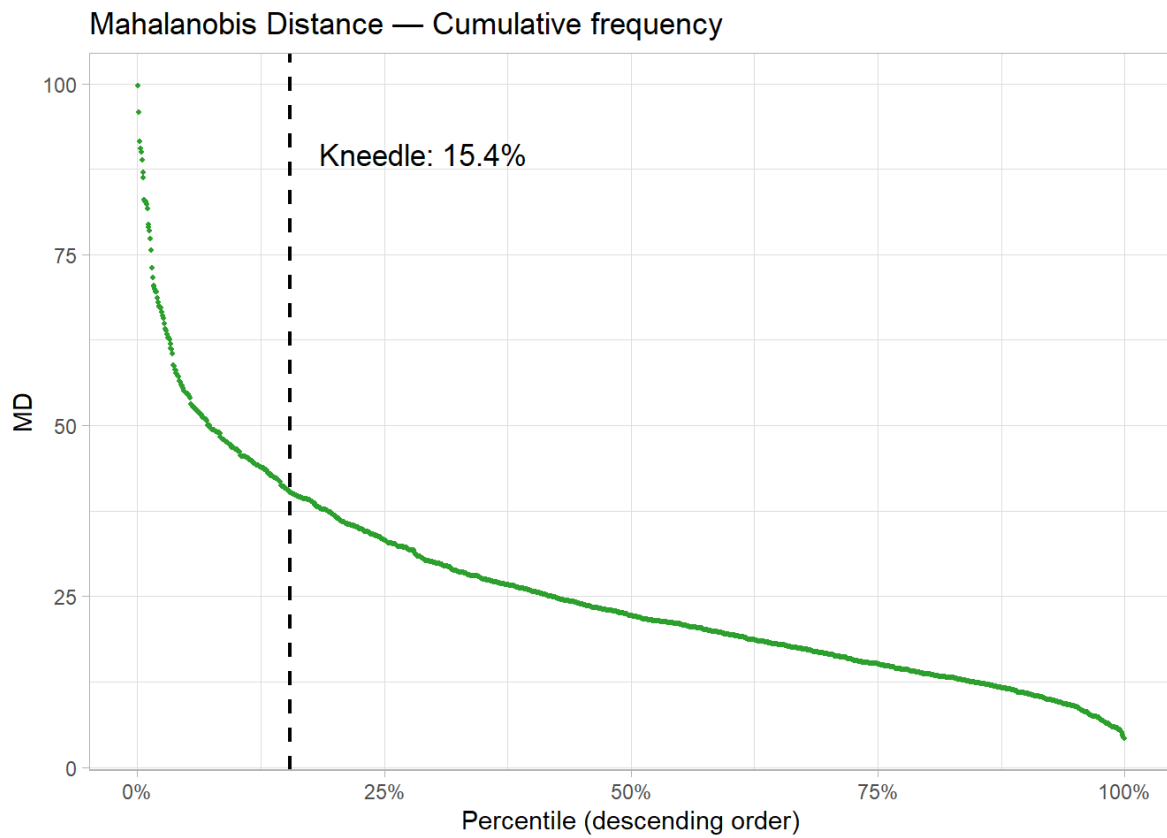
Source: [Post-hoc C/IER Indicators](#)

4.3.2 Longstring



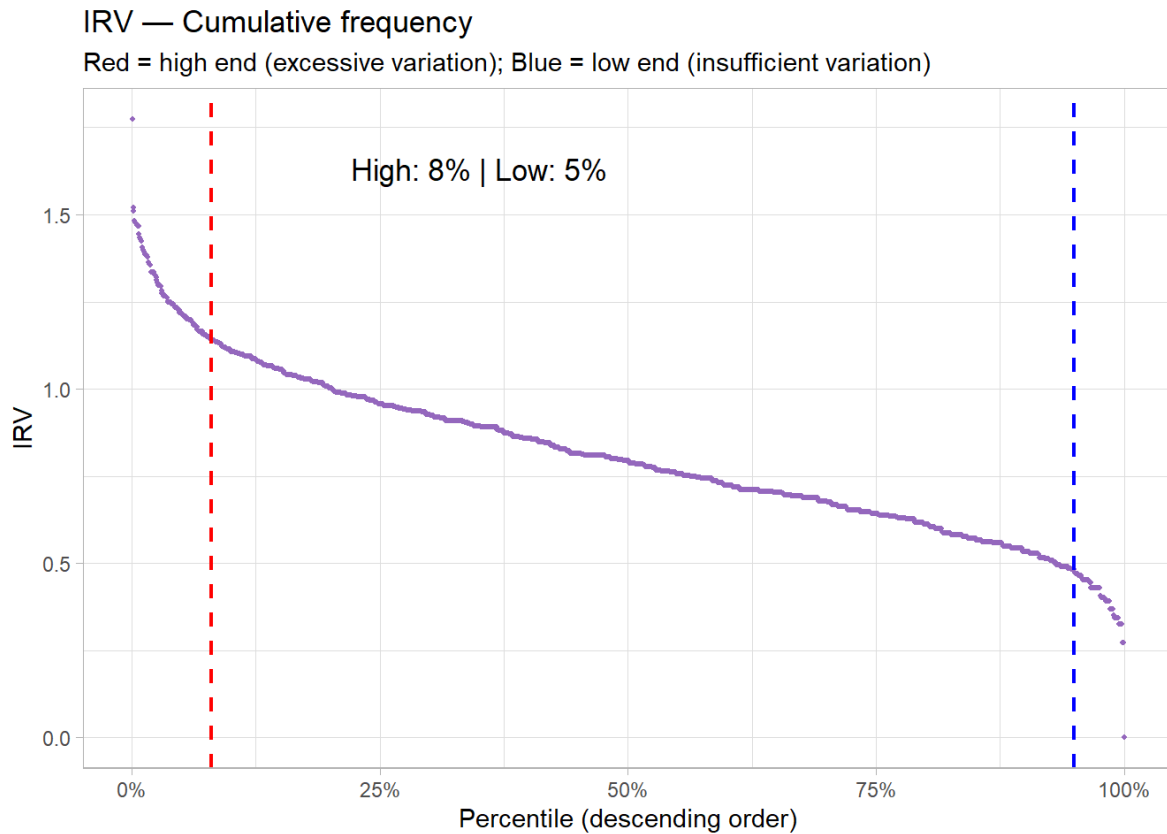
Source: [Post-hoc C/IER Indicators](#)

4.3.3 Mahalanobis Distance



Source: [Post-hoc C/IER Indicators](#)

4.3.4 IRV



Source: [Post-hoc C/IER Indicators](#)

4.4 C/IER Flags

For each indicator, a `_FLAG` variable is created (1 = flagged, 0 = not flagged) based on the Kneedle cut-off points.

Note: Post-hoc indicators are **diagnostic**, they do not trigger automatic exclusion in this phase. The composite flag array will be used for inspection and evaluation through the `whoqol_screened.csv` file.

Table 3: Summary of flagged C/IER respondents

Indicator	N flagged	% flagged
LazR_FLAG	152	11.7
Longstr_FLAG	196	15.1

Indicator	N flagged	% flagged
MD_FLAG	201	15.5
IRV_high_FLAG	108	8.3
IRV_low_FLAG	65	5.0

Source: [Post-hoc C/IER Indicators](#)

- Biemann, T., Koch-Bayram, I. F., Meier-Barthold, M., & Aguinis, H. (2025). Using Markov Chains to Detect Careless Responding in Survey Research. *Organizational Research Methods*, 28(4), 543–568. <https://doi.org/10.1177/10944281251334778>
- Curran, P. G. (2016). Methods for the detection of carelessly invalid responses in survey data. *Journal of Experimental Social Psychology*, 66, 4–19. <https://doi.org/10.1016/j.jesp.2015.07.006>
- Hong, M., Steedle, J. T., & Cheng, Y. (2020). Methods of Detecting Insufficient Effort Responding: Comparisons and Practical Recommendations. *Educ Psychol Meas*, 80(2), 312–345. <https://doi.org/10.1177/0013164419865316>
- Huang, J. L., Wang, Z., Huang, R., Wu, D., & Shi, H. (2026). *Insufficient Effort Responding in Management Research: A Critical Review and Future Directions*. 52(1). <https://doi.org/10.1177/01492063251330268>